

Free mechanical vibrations

$$\overset{\text{mass}}{[\text{inertia}] \underset{m}{\frac{d^2 y}{dt^2}}} + \underset{b}{[\text{damping}] \frac{dy}{dt}} + \overset{\text{spring}}{[\text{stiffness}] \underset{k}{y}} = F_{\text{ext}}$$

1° Undamped, free case ($b=0, F_{\text{ext}}=0$)

$$m y'' + k y = 0, \quad \omega = \sqrt{\frac{k}{m}} \Rightarrow$$

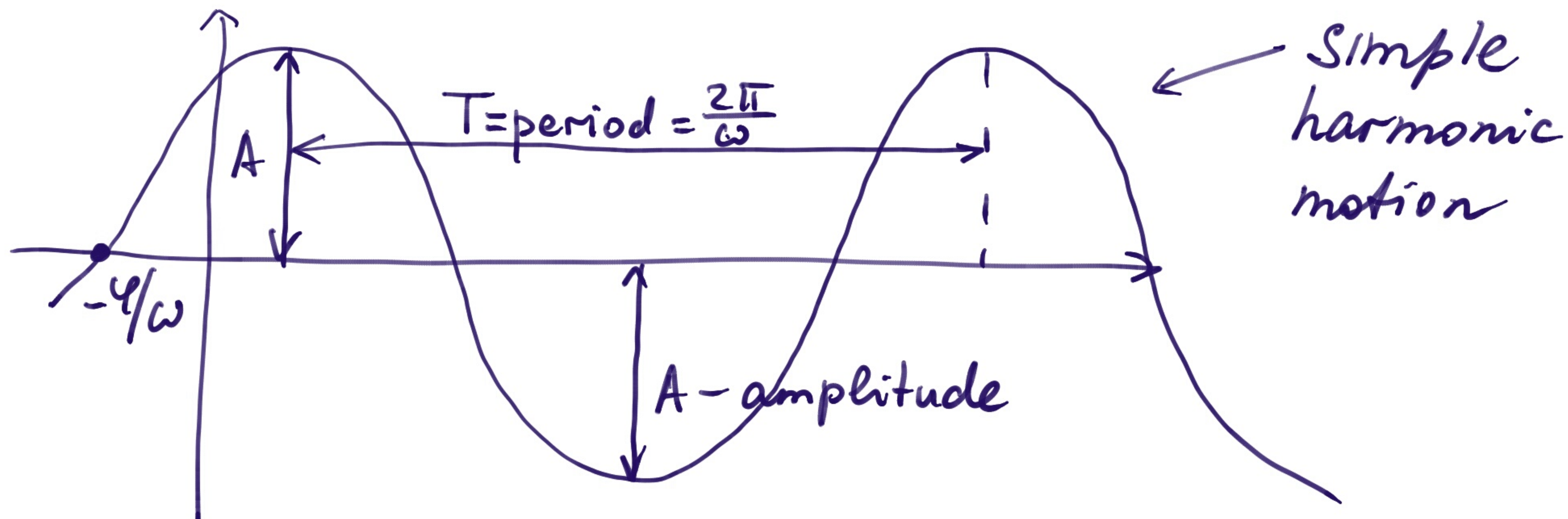
$$\Rightarrow y'' + \omega^2 y = 0 \Rightarrow \lambda^2 + \omega^2 = 0 \Rightarrow \lambda_{1,2} = \pm \omega i$$

$$\Rightarrow y = c_1 \cos \omega t + c_2 \sin \omega t, \quad \boxed{y = A \sin(\omega t + \varphi)}$$

$$A \sin(\omega t + \varphi) = \underbrace{A \cos \varphi}_{c_2} \sin \omega t + \underbrace{A \sin \varphi}_{c_1} \cos \omega t$$

$$A = \sqrt{c_1^2 + c_2^2}$$

$$\frac{c_1}{c_2} = \frac{\sin \varphi}{\cos \varphi} = \tan \varphi \Rightarrow \varphi = \arctan \frac{c_1}{c_2}$$



$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad \leftarrow \text{period}$$

$\omega \leftarrow$ angular frequency

natural frequency $\nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{\omega}{2\pi}$

ϕ - phase angle

Underdamped motion ($b^2 < 4mk$)

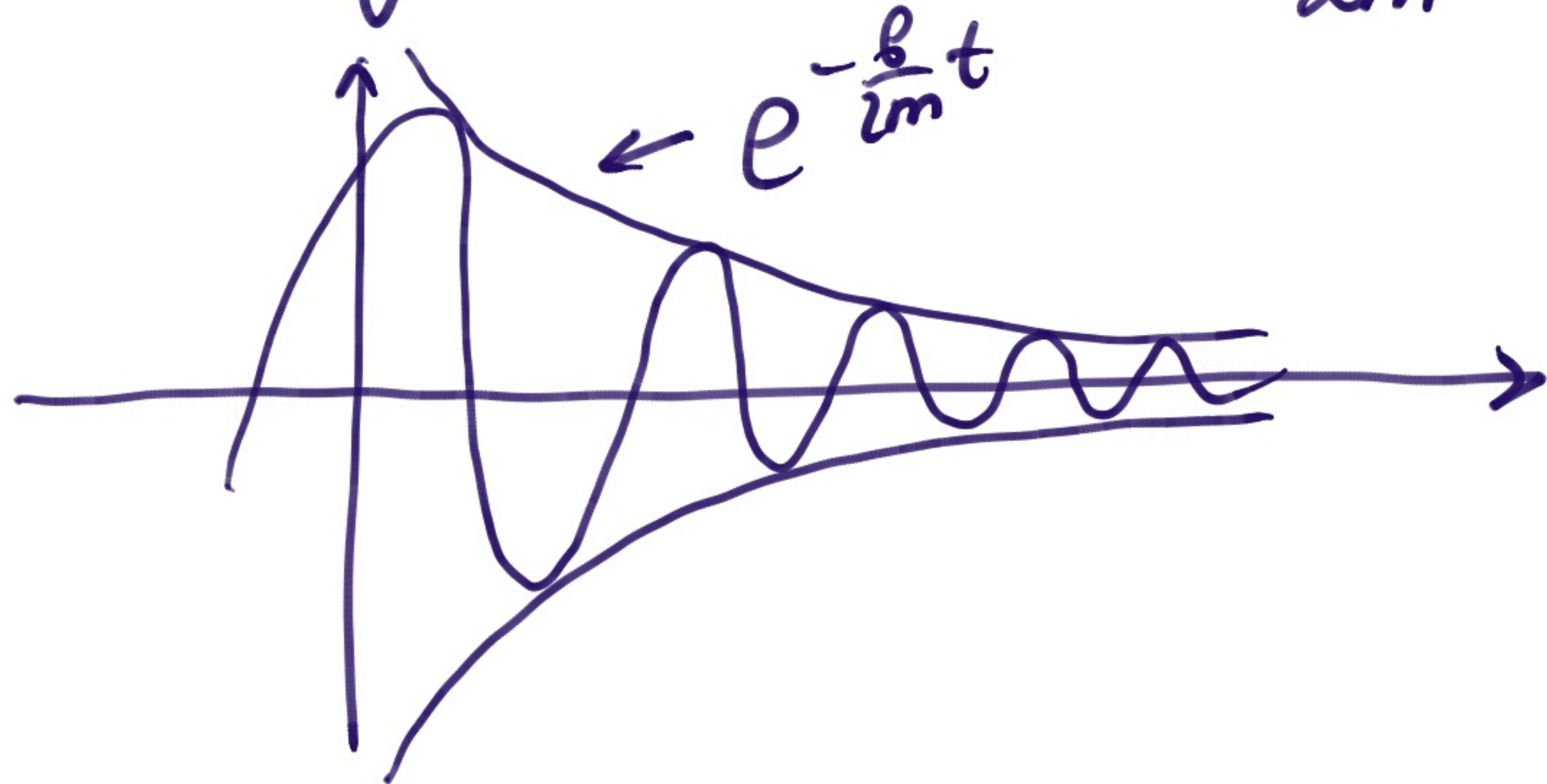
$$my'' + by' + ky = 0$$

$$mr^2 + br + k = 0$$

$$D = b^2 - 4mk < 0$$

$$r_{1,2} = \frac{-b \pm i\sqrt{4mk - b^2}}{2m} = -\frac{b}{2m} + i\frac{\sqrt{4mk - b^2}}{2m}$$

$$y = C_1 e^{-\frac{b}{2m}t} \cos \frac{\sqrt{4mk - b^2}}{2m}t + C_2 e^{-\frac{b}{2m}t} \sin \frac{\sqrt{4mk - b^2}}{2m}t$$



$$y = A e^{-\frac{b}{2m}t} \sin(\beta t + \varphi)$$

$$T = \frac{2\pi}{\beta} \leftarrow \text{quasiperiod}$$

$$\nu = \frac{1}{T} \leftarrow \text{quasifrequency}$$

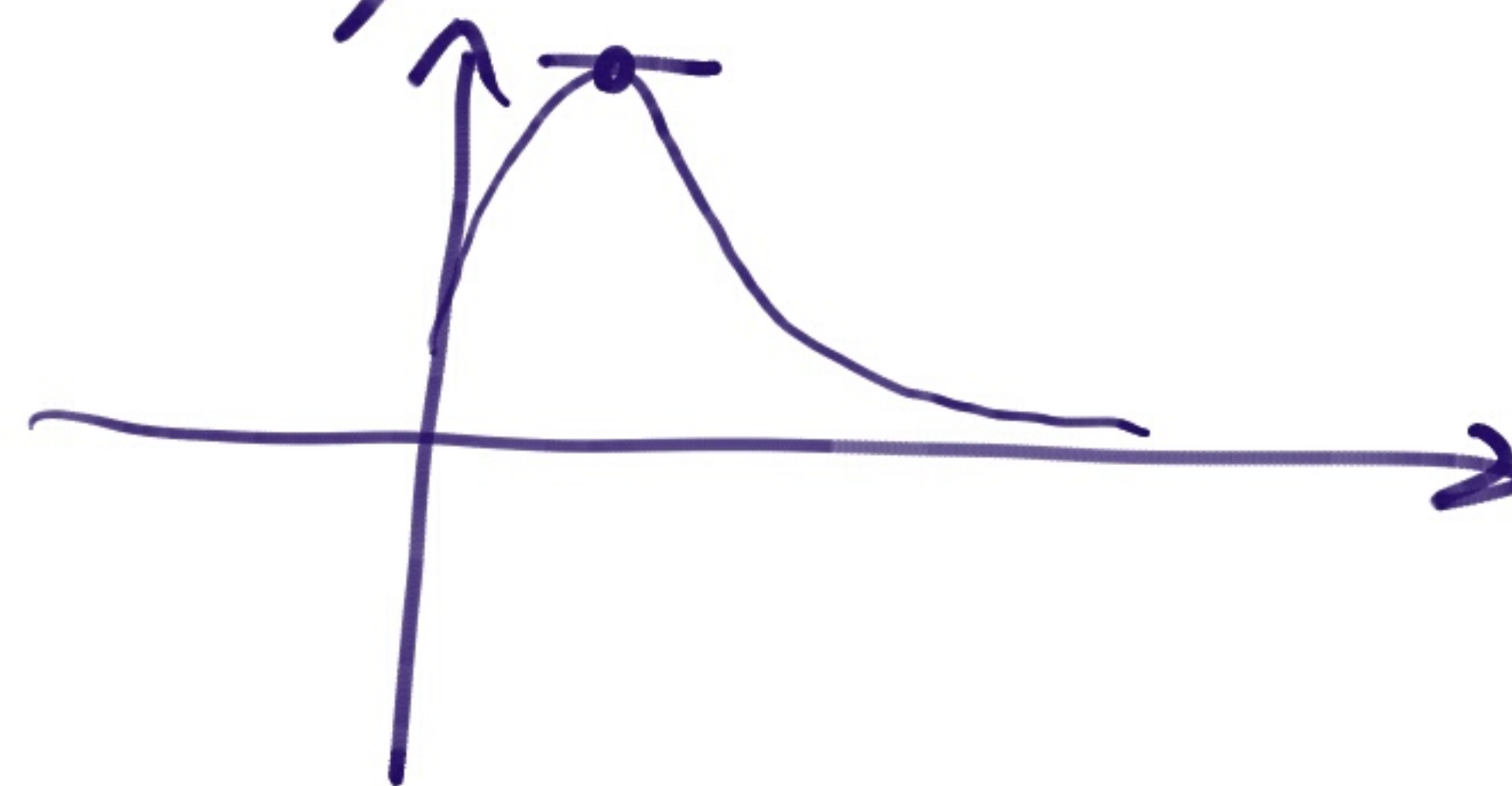
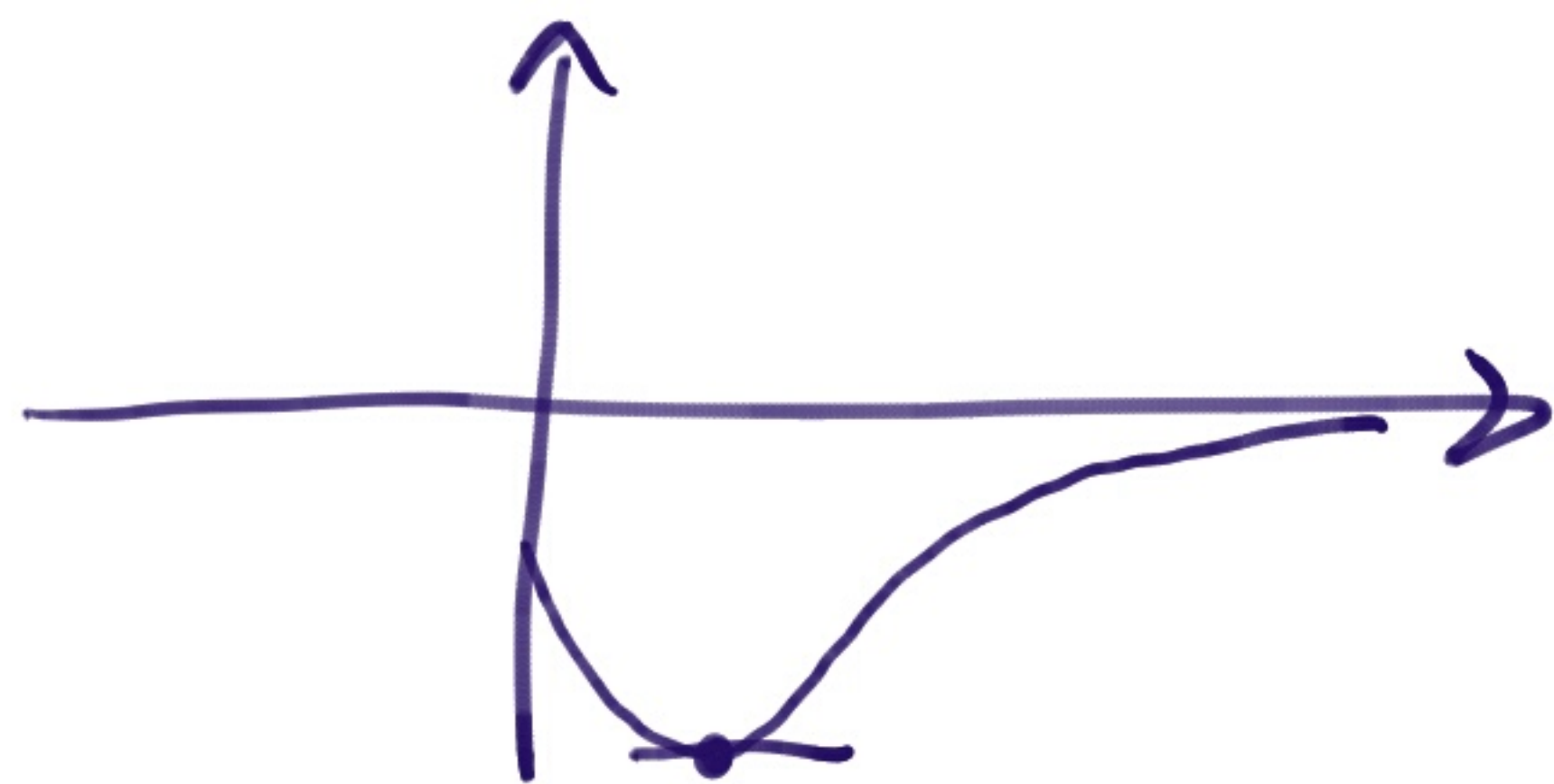
Critically damped motion ($b^2 = 4mk$)

$$r_1 = r_2 = -\frac{b}{2m}$$

$$y(t) = C_1 e^{-\frac{b}{2m}t} + C_2 t e^{-\frac{b}{2m}t} = (C_1 + C_2 t) e^{-\frac{b}{2m}t}$$

$$\lim_{t \rightarrow \infty} \frac{C_1 + C_2 t}{e^{\frac{b}{2m}t}} \stackrel{L'H}{=} \lim_{t \rightarrow \infty} \frac{C_2}{\frac{b}{2m} e^{\frac{b}{2m}t}} = 0$$

$$y'(t) = \left(-\frac{b C_1}{2m} + C_2 - \frac{b}{2m} C_2 t\right) e^{-\frac{b}{2m}t} = 0 \Rightarrow \text{max or min at one point}$$



Overdamped motion ($b^2 > 4mk$)

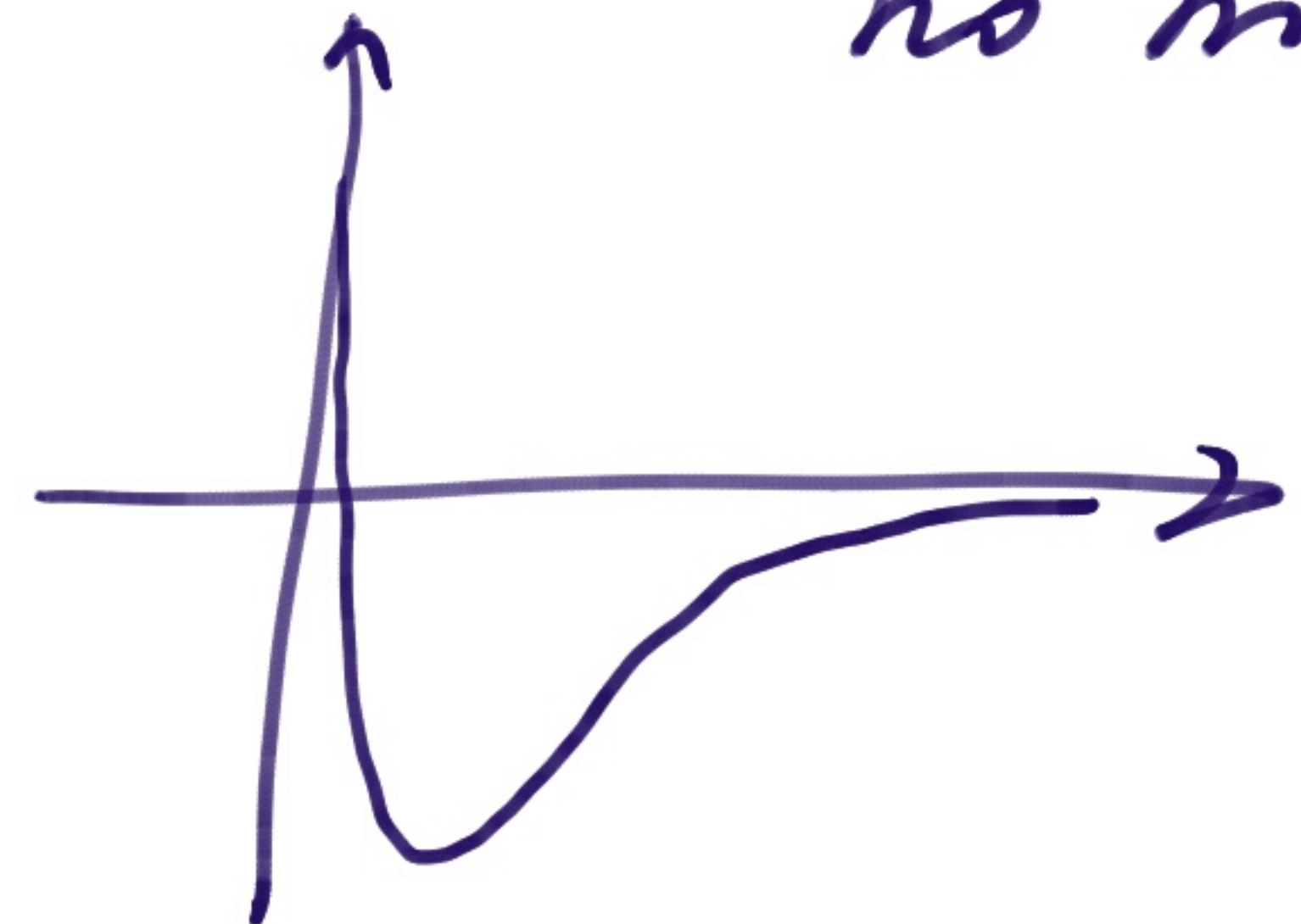
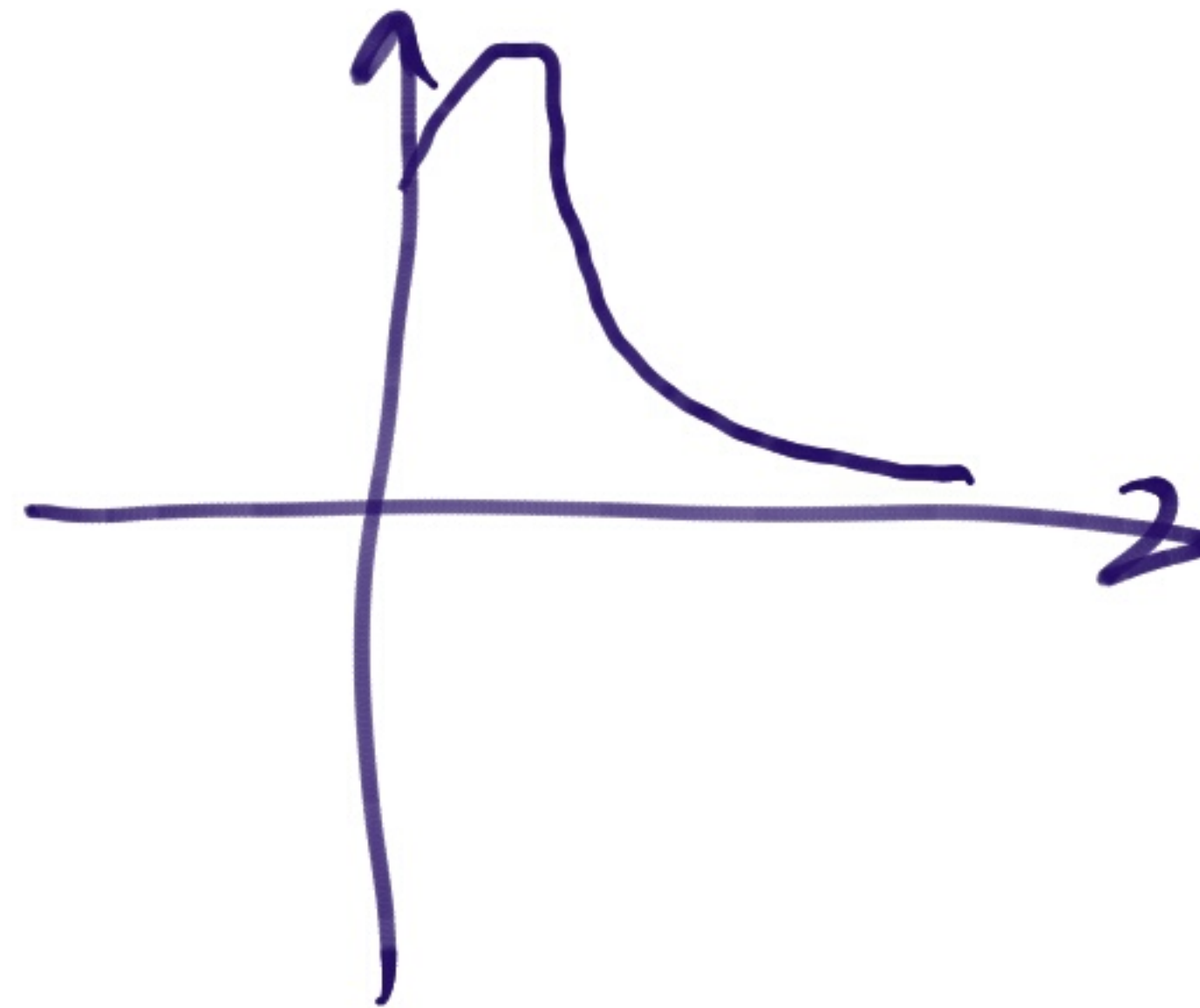
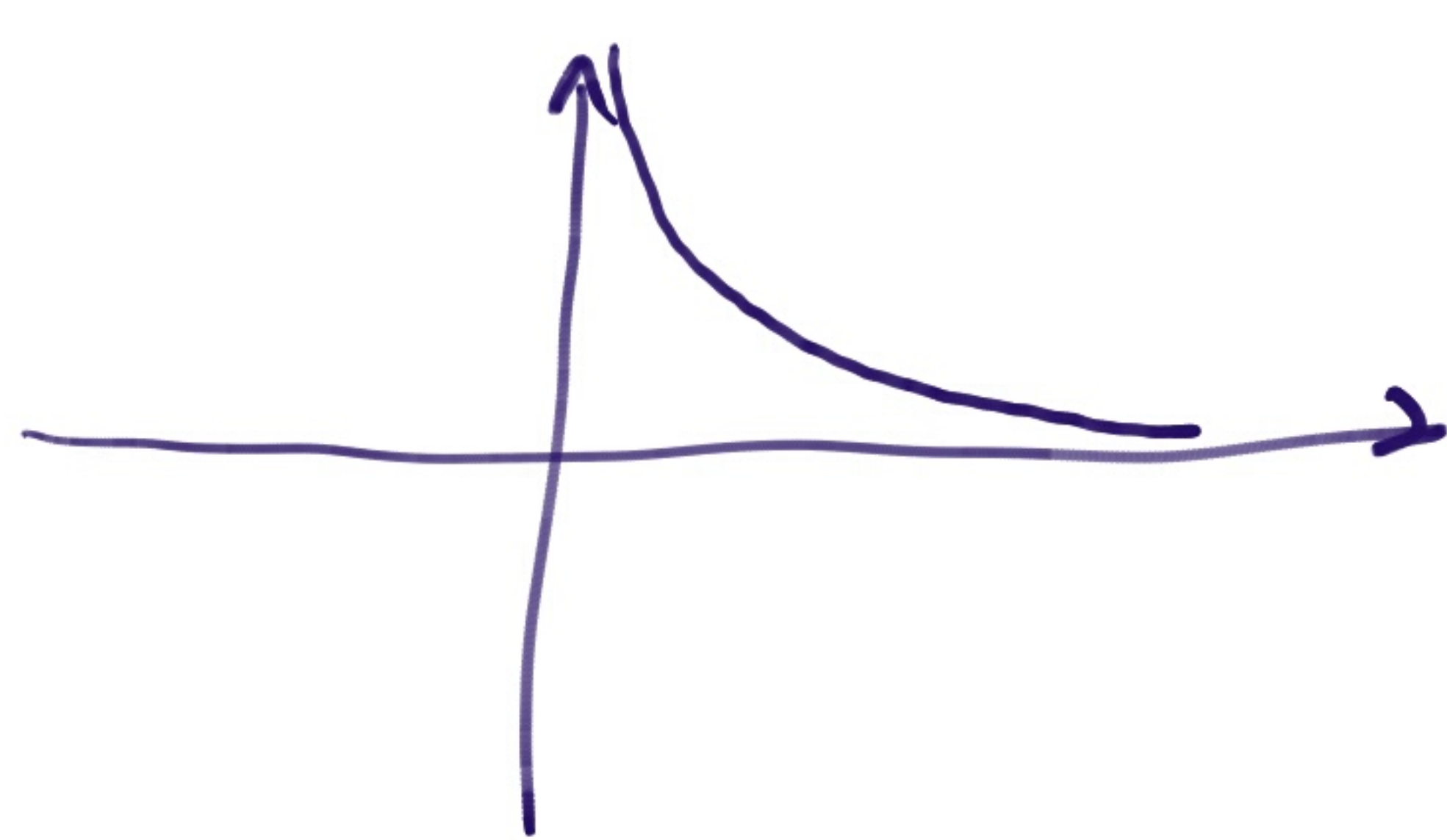
$$r_1 \neq r_2 \Rightarrow r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

$$y = c_1 e^{r_1 t} + c_2 e^{r_2 t} \xrightarrow[t \rightarrow \infty]{} 0$$

$$y' = c_1 r_1 e^{r_1 t} + c_2 r_2 e^{r_2 t} =$$

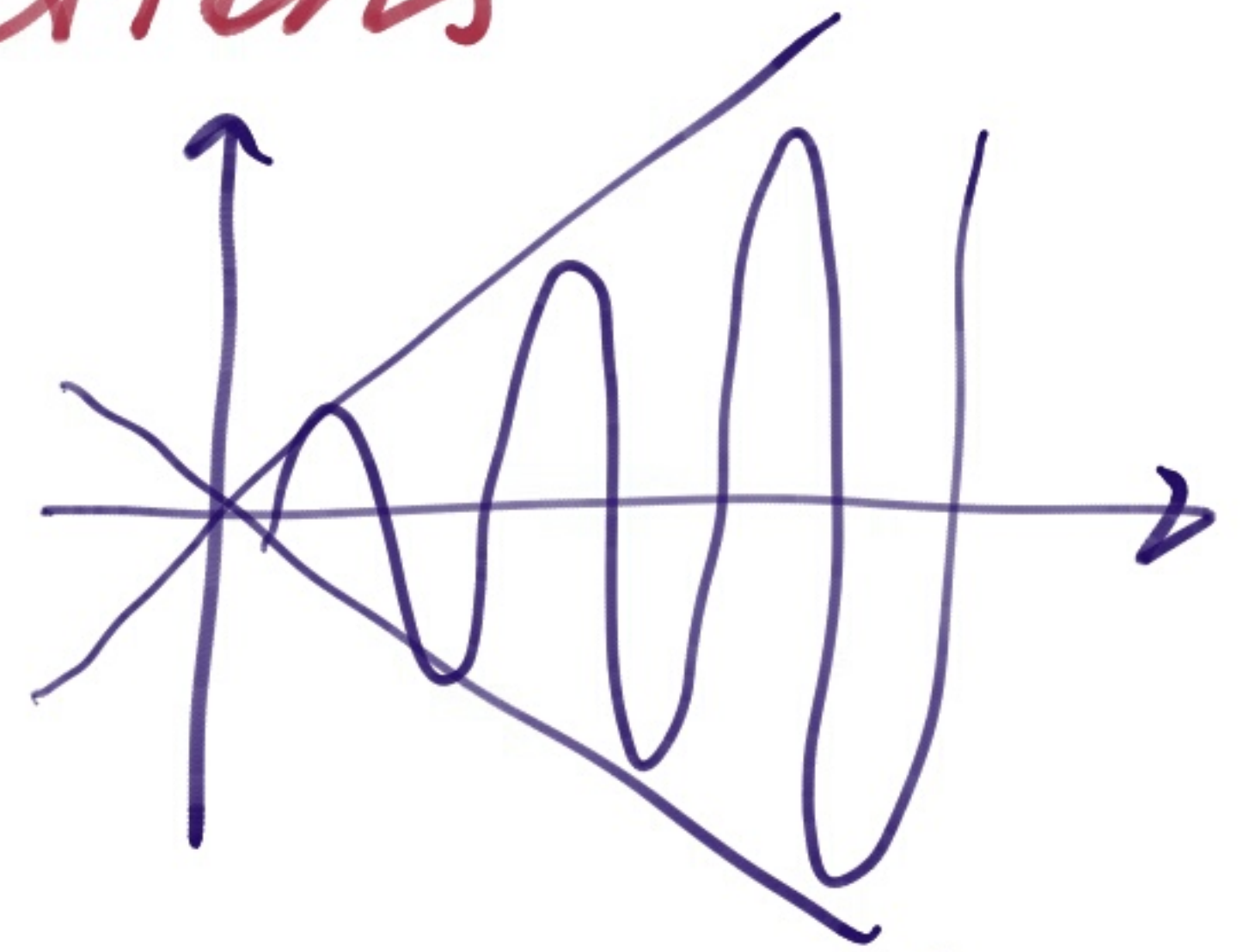
$$= e^{r_1 t} (c_1 r_1 + c_2 r_2 e^{(r_2 - r_1)t}) = 0$$

only one
local max
or min
or no max
no min



Forced mechanical vibrations

$$my'' + by' + ky = F_{\text{ext}}(t)$$



$$b=0$$

$$y'' + \omega^2 y = \cos \gamma t$$

$$1^\circ \quad y_H = A \sin(\omega t + \varphi) = c_1 \cos \omega t + c_2 \sin \omega t$$

$$2^\circ - y_P = B \cos \gamma t + D \sin \gamma t = A_1 \sin(\gamma t + \varphi_1) \quad \left(\begin{array}{l} \text{if} \\ \omega \neq \gamma \end{array} \right)$$

$$y = A \sin(\omega t + \varphi) + A_1 \sin(\gamma t + \varphi_1) \leftarrow \text{bounded}$$

$$- y_P = (B \cos \gamma t + D \sin \gamma t) t \quad (\text{if } \omega = \gamma)$$

$$y = A \sin(\omega t + \varphi) + \underbrace{(t A_1 \sin(\gamma t + \varphi_1))}_{t \rightarrow \infty} \rightarrow \infty \quad \leftarrow \boxed{\text{resonance}}$$